

24-Sept

Quantum Formalism - Whapping things up (Ch 3)

3.6 - Vectors in Hilbert Space

State in QM $|\psi\rangle$ is a vector in a linear vector space
operators - linear transformation in State Space.

As an example - Suppose we have a 2-State system (qubit)

$$|\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Let's write H in this basis as

$$H_{00} = \langle \psi_0 | H | \psi_0 \rangle = E_0$$

$$H_{10} = H_{01}^* = \langle \psi_1 | H | \psi_0 \rangle = g \text{ (may be complex)}$$

$$H_{11} = E_1$$

$$H = \begin{bmatrix} E_0 & g \\ g & E_1 \end{bmatrix}$$

I can define ~~the~~ operators in the $\{|\psi_0\rangle, |\psi_1\rangle\}$ basis.

$$\text{eg. } \sigma_x = \frac{1}{\hbar} [|\psi_1\rangle\langle\psi_0| + |\psi_0\rangle\langle\psi_1|]$$

$$\sigma_y = i [|\psi_1\rangle\langle\psi_0| - |\psi_0\rangle\langle\psi_1|]$$

$$\sigma_z = [|\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|]$$

$$\sigma_0 = [|\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1|]$$

for reasons that
will become more
obvious when we
talk about Spin $\frac{1}{2}$
Systems

I can write.

$$H = a_0 \sigma_0 + a_1 \sigma_x + a_2 \sigma_y + a_3 \sigma_z$$

$$\Delta = \frac{\epsilon_0 - \epsilon_1}{2} = \frac{\Delta}{2}$$

$$\frac{\epsilon_0 + \epsilon_1}{2} = a_0; \quad a_1 = \frac{\Delta}{2}; \quad a_2 = \text{Im}(g), \quad a_3 = \text{Re}(g)$$

let's define a transformation that brings H into Diagonal form.

w/o loss of generality $H = \frac{\Delta}{2} \sigma_z + g \sigma_x$ and take g to be real.

Define a rotation matrix $T(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

$$T^\dagger T = \mathbb{1}$$

Now take $\tilde{H} = T^\dagger H T$ and define $|\psi_+\rangle = T^\dagger |\psi_1\rangle$

$$|\psi_-\rangle = T^\dagger |\psi_2\rangle$$

$$\tilde{H} = \begin{bmatrix} \frac{1}{2} \Delta \cos(2\theta) - g \sin(2\theta) & g(\cos(2\theta) + \Delta \cos(\theta) \sin(\theta)) \\ g \cos(2\theta) + \Delta \cos(\theta) \sin(\theta) & -\frac{1}{2} \Delta \cos(2\theta) + g \sin(2\theta) \end{bmatrix}$$

find θ so that $\tilde{H}_{+-} = 0$

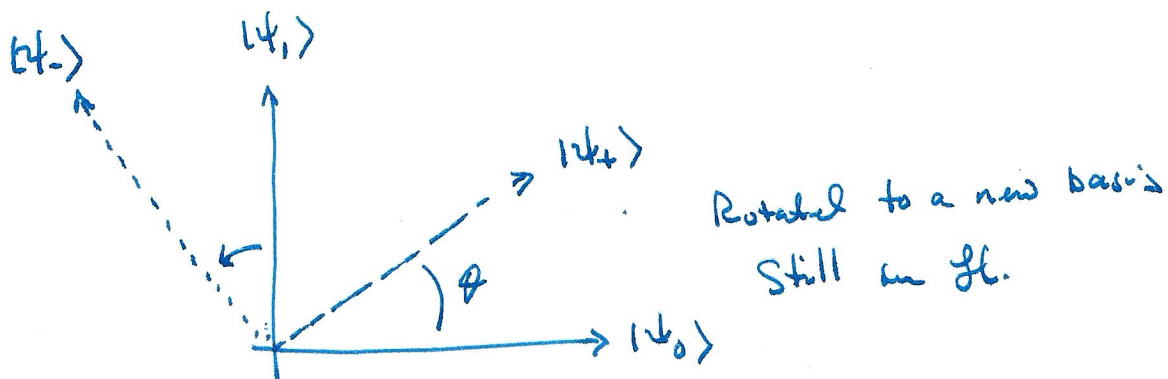
$$g \cos(2\theta) + \Delta \cos \theta \sin \theta = 0$$

$$\Rightarrow g \cos(2\theta) = -\Delta \cos \theta \sin \theta$$

$$\frac{-g \cos(2\theta)}{\Delta \cos \theta \sin \theta} = 1 = \frac{-2g \cot 2\theta}{\Delta}$$

$$\boxed{\cot \frac{-\Delta}{2g} = \cot(2\theta)} \rightarrow \text{gives angle}$$

$$\tan(2\theta) = \frac{-2g}{\Delta} \xrightarrow{\text{coupling}} \frac{g}{\Delta}$$



$$\begin{aligned} \frac{1}{2} \left. \begin{aligned} E_+ &= \frac{1}{2} \Delta \cos(2\theta) - g \sin(2\theta) \\ E_- &= -\frac{1}{2} \Delta \cos(2\theta) + g \sin(2\theta) \end{aligned} \right\} \text{New eigenvalues of } H. \end{aligned}$$

$$E_{\pm} = \pm \frac{1}{2} \sqrt{4g^2 + \Delta^2}.$$

Suppose we prepare an initial state in $|\psi_0\rangle$ and allow it to evolve in time according to S.E.

$$|\psi_0\rangle = \cos\theta |\psi_+\rangle + \sin\theta |\psi_-\rangle$$

$$\begin{aligned} |\psi(t)\rangle &= \exp\left[-i \frac{\hat{H} t}{\hbar}\right] |\psi_0\rangle \\ &= \cos\theta e^{-i E_+ t / \hbar} |\psi_+\rangle + \sin\theta e^{-i E_- t / \hbar} |\psi_-\rangle \end{aligned}$$

to make the on board calc easier - suppose $\Delta = 0 \neq g$ so
 $\cos\theta = 1/\sqrt{2}$ $\sin\theta = 1/\sqrt{2}$ $\theta = \pi/4$, $E_{\pm} = \pm \frac{1}{2} \sqrt{4g^2} = \pm g$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left[\exp\left[\frac{-ig t}{\hbar}\right] |\psi_+\rangle + \exp\left[\frac{+ig t}{\hbar}\right] |\psi_-\rangle \right]$$

Prob to remain (or find system) in $|\psi_0\rangle$ at time t ?

$$P_0(t) = |\langle \psi_0 | \psi(t) \rangle|^2$$

$$\begin{aligned} \langle \psi_0 | \psi(t) \rangle &= \frac{1}{\sqrt{2}} [\langle \psi_+ | + \langle \psi_- |] \left[\frac{1}{\sqrt{2}} \right] [e^{-i\frac{gt}{\hbar}} |\psi_+\rangle + e^{+i\frac{gt}{\hbar}} |\psi_-\rangle] \\ &= \frac{1}{2} [e^{-igt/\hbar} + e^{+igt/\hbar}] = \cos\left(\frac{gt}{\hbar}\right) \end{aligned}$$

$$\underline{P_0(t) = \cos^2(gt/\hbar)} \quad \frac{g}{\hbar} = \Omega \text{ - Rabi Freq.}$$

Similarly $P_1 = 1 - P_0(t)$ Since $P_0 + P_1 = 1$

- Coherent oscillation between $|\psi_0\rangle$ & $|\psi_1\rangle$ states

Some operator properties and identities.

$$1. (\hat{Q} + \hat{R})|\psi\rangle = \hat{Q}|\psi\rangle + \hat{R}|\psi\rangle$$

$$2.) \hat{Q}\hat{R}|\psi\rangle = \hat{Q}(\hat{R}|\psi\rangle) = \hat{Q}|\psi'\rangle$$

$$\text{Note } \hat{Q}\hat{R}|\psi\rangle \neq \hat{R}\hat{Q}|\psi\rangle \text{ if } [\hat{Q}, \hat{R}] \neq 0 \quad !!$$

Some properties

$f(\hat{Q})$ - Need to do power series expansion in terms of op.

~~$f(\hat{Q}) = 1 + \hat{Q} + \dots$~~

$$\text{eg } e^{\hat{Q}} = 1 + \hat{Q} + \frac{1}{2!} \hat{Q}^2 + \frac{1}{3!} \hat{Q}^3 + \dots$$

$$\frac{1}{1-\hat{Q}} = 1 + \hat{Q} + \hat{Q}^2 + \hat{Q}^3 + \dots$$

$$\ln(1+\hat{Q}) = \hat{Q} - \frac{1}{2}\hat{Q}^2 + \frac{1}{3}\hat{Q}^3 - \frac{1}{4}\hat{Q}^4 + \dots$$

HW4

Prove these identities

$$\text{Be careful: } e^{\hat{A}+\hat{B}} \neq e^{\hat{A}}e^{\hat{B}} \quad \text{if } [\hat{A}, \hat{B}] \neq 0$$

And $[\hat{A}, \hat{B}] \neq 0$

$$\text{Suppose } e^{\hat{A}}e^{\hat{B}} = e^{\hat{C}} = 1 + \hat{C} + \frac{1}{2}\hat{C}^2 + \dots$$

Suppose

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{\hat{C}} \quad \ln(e^{\hat{A}}e^{\hat{B}}) = \hat{A} + \hat{B} + \frac{1}{2}[\hat{A}, \hat{B}] + \dots$$

$\therefore$ the compensating exponent

$$\text{begins } \hat{C} = -\frac{1}{2}[\hat{A}, \hat{B}] + \dots$$

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A,B]} + \dots$$

Special case: $[A, [A, B]] = [B, [A, B]] = 0$ - Series terminates

~~Useful approx for $U = e^S$~~

Often we may want to write Quantum Ops As unitary transformations $U = e^S$ when S is "Anti Hermitian"

$$S^\dagger = -S$$

$$U^\dagger = e^{-S}$$

Very often we'll want to write

$$\tilde{A} = U^\dagger A U = e^{-S} A e^{+S}$$

$$= A + i[A, S] + \frac{1}{2!} [[A, S], S] + \frac{1}{3!} [[[A, S], S], S] + \dots$$

Suppose we write

$$H = H_0 + W$$

H_0 - diagonal

W - not diagonal.

$$\langle m | W | n \rangle \neq 0$$

$$\langle m | W | m \rangle = 0$$

$$\tilde{H} = H_0 + \frac{1}{1!} [H_0, S] + \frac{1}{2!} [[H_0, S], S] + \dots$$

$$+ W + \frac{1}{1!} [W, S] + \frac{1}{2!} [[W, S], S] + \dots$$

let's pick in S so that $\tilde{H}$ is now Diagonal.

Simpler is to require $[H_0, S] = -W$.

If we require this then

$$\tilde{H} = H_0 + \left(\frac{1}{1!} - \frac{1}{2!}\right)[W, S] + \left(\frac{1}{2!} - \frac{1}{3!}\right)[[W, S], S] + \dots$$

going back to our 2-state problem

$$H_0 = \frac{\Delta}{2} \sigma_z \quad W = g \sigma_x$$

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$$[H_0, S] = -W$$

$$\frac{\Delta}{2} [\sigma_z, S] = -g \sigma_x$$

$$\cancel{[S, \sigma_y]} = [\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k$$

$$[\sigma_x, \sigma_y] = 2i \sigma_z \Rightarrow \underline{[\sigma_y, \sigma_z] = 2i \sigma_x}$$

$$\cancel{\frac{1}{2} [\sigma_z, \sigma_y] = \sigma_x}$$

$$\Rightarrow S = a \sigma_y \Rightarrow \underline{a = -i g / \Delta}$$

$$\therefore \tilde{H} = e^{+i \frac{g}{\Delta} \sigma_y} H_0 e^{-i \frac{g}{\Delta} \sigma_y} \quad \text{Rotation about } y$$

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